

We're going to build a classification problem in three dimensions using a degree 2 polynomial feature map.

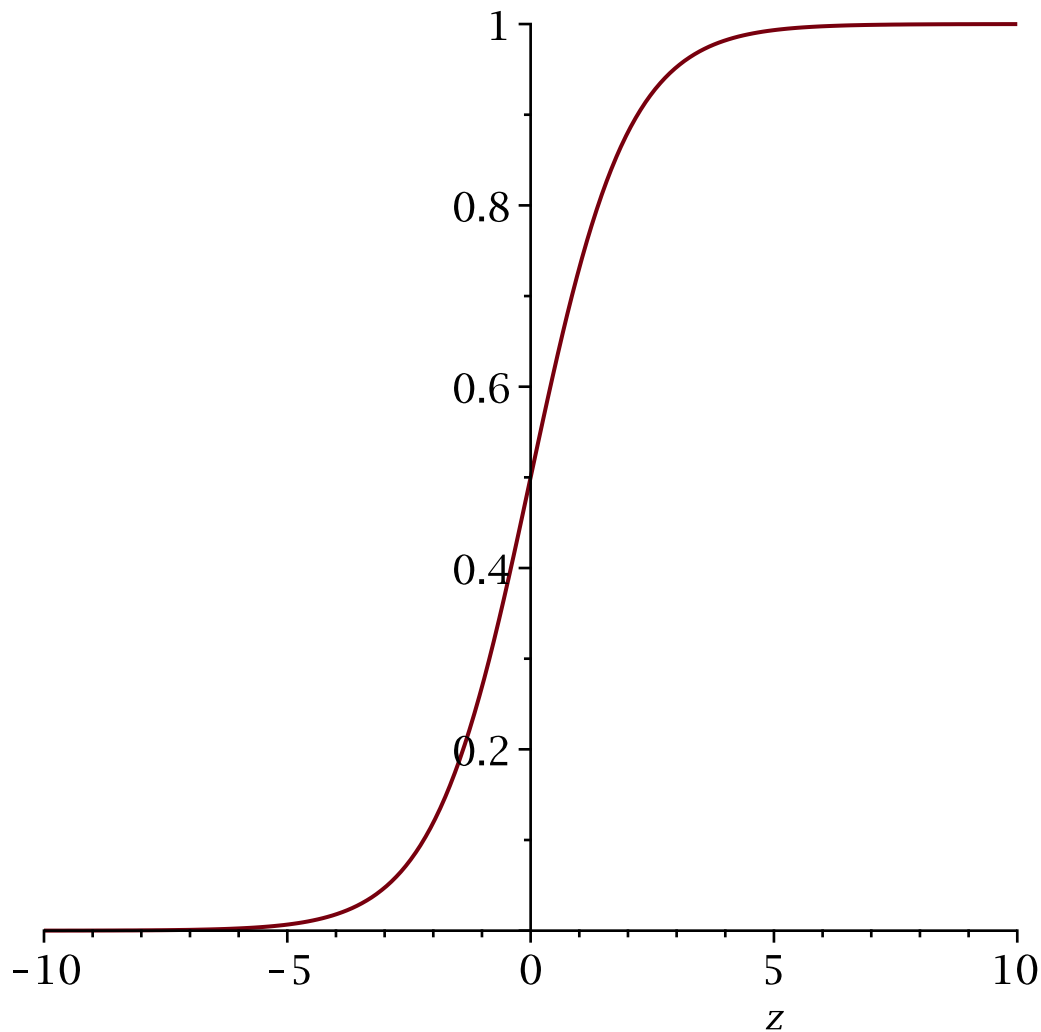
We'll start by initializing a 3-d manifold in which the original datapoints are given.

```
> DGEnvironment[Manifold]([x,y,z],M)  
Manifold: M
```

(2)

Here we'll apply logistic regression, which assigns a probability >0.5 when $z > 0$ and probability <0.5 when $z < 0$:

```
M > sigmoid:=1/(1+exp(-z)):  
plot(sigmoid);
```



Now the trick is that "z" is going to be $x^2 + y^2/2 - yz + z^2/2$: if this is positive at a point then the point belongs to class 1 and if this is negative then the point belongs

to class 0.

```
M > zee:=x^2+y^2/2-y*z+z^2/2;
```

$$zee := x^2 + \frac{1}{2} y^2 - yz + \frac{1}{2} z^2 \quad (3)$$

We're going to build a classification problem

```
M > f:=eval(sigmoid,z=zee);
```

$$f := \frac{1}{1 + e^{-x^2 - \frac{1}{2} y^2 + yz - \frac{1}{2} z^2}} \quad (4)$$

One way to do this is to move to a feature space in which "z" is (hopefully) a hyperplane.

Our feature map into 6 dimensions will be $x_1=x^2$, $x_2=\sqrt{2}*x*y$, $x_3=\sqrt{2}*x*z$, $x_4=y^2$, $x_5=\sqrt{2}*y*z$, $x_6=z^2$.

If you get to reading about kernels and feature maps in machine learning, here is the kernel for this feature map:

```
M > kernel:=(x1*z1+x2*z2+x3*z3)^2
```

$$kernel := (x_1 z_1 + x_2 z_2 + x_3 z_3)^2 \quad (5)$$

```
M > expand(kernel)
```

$$x_1^2 z_1^2 + 2 x_1 x_2 z_1 z_2 + 2 x_1 x_3 z_1 z_3 + x_2^2 z_2^2 + 2 x_2 x_3 z_2 z_3 + x_3^2 z_3^2 \quad (6)$$

Calculating the kernel is the same as computing the (Euclidean) inner product in the feature space:

```
M > Matrix([x1^2,sqrt(2)*x1*x2,sqrt(2)*x1*x3,x2^2,sqrt(2)*x2*x3,
x3^2]).LinearAlgebra:-IdentityMatrix(6).Vector([z1^2,sqrt(2)*
z1*z2,sqrt(2)*z1*z3,z2^2,sqrt(2)*z2*z3,z3^2]) ;
```

$$\left[x_1^2 z_1^2 + 2 x_1 x_2 z_1 z_2 + 2 x_1 x_3 z_1 z_3 + x_2^2 z_2^2 + 2 x_2 x_3 z_2 z_3 + x_3^2 z_3^2 \right] \quad (7)$$

Now we will build our 6-d feature space.

```
M > DGEnvironment[Manifold]([x1,x2,x3,x4,x5,x6],M6)
```

$$\text{Manifold: } M_6 \quad (8)$$

With the Euclidean metric tensor.

```
M6 > gM6:=convert(LinearAlgebra:-IdentityMatrix(6),DGtensor,[
```

$$["cov_bas", "cov_bas"], [[]])$$

$$gM6 := dx1 \otimes dx1 + dx2 \otimes dx2 + dx3 \otimes dx3 + dx4 \otimes dx4 + dx5 \otimes dx5 + dx6 \otimes dx6 \quad (9)$$

Here is the feature map

$$M6 > eqns := [x1 = x^2, x2 = \sqrt{2}xy, x3 = \sqrt{2}xz, x4 = y^2, x5 = \sqrt{2}yz, x6 = z^2];$$

$$eqns := [x1 = x^2, x2 = \sqrt{2}xy, x3 = \sqrt{2}xz, x4 = y^2, x5 = \sqrt{2}yz, x6 = z^2] \quad (10)$$

$$M6 > Phi := Transformation(M, M6, eqns)$$

$$\Phi := x1 = x^2, x2 = \sqrt{2}xy, x3 = \sqrt{2}xz, x4 = y^2, x5 = \sqrt{2}yz, x6 = z^2 \quad (11)$$

And here is the metric tensor in the original (x,y) coordinates induced by this feature map.

$$M > g := Pullback(Phi, gM6);$$

$$g := (4x^2 + 2y^2 + 2z^2) dx \otimes dx + 2yx dx \otimes dy + 2zx dx \otimes dz + 2yx dy \otimes dx + (2x^2 + 4y^2 + 2z^2) dy \otimes dy + 2yz dy \otimes dz + 2zx dz \otimes dx + 2yz dz \otimes dy + (2x^2 + 2y^2 + 4z^2) dz \otimes dz \quad (12)$$

I thought for a minute about using the determinant of g to generate points, but I think we'll have to stick with a more simple data generation tool for now.

$$M > detg := LinearAlgebra:-Determinant(convert(g, DGMatrix));$$

$$detg := 16x^6 + 48x^4y^2 + 48x^4z^2 + 48x^2y^4 + 96x^2y^2z^2 + 48x^2z^4 + 16y^6 + 48y^4z^2 + 48y^2z^4 + 16z^6 \quad (13)$$

Now we calculate the infinitesimal isometries.

$$M > kv := KillingVectors(g);$$

$$M > nops(kv);$$

$$3 \quad (14)$$

$$M > kv;$$

$$\left[-\frac{z}{2} \partial_y + \frac{y}{2} \partial_z, -\frac{z}{2} \partial_x + \frac{x}{2} \partial_z, -\frac{y}{2} \partial_x + \frac{x}{2} \partial_y \right] \quad (15)$$

In the next few lines, I just wanted to be sure that there really were only 3 isometries. Maple's pdsolve is great, but who's to say it didn't miss solutions? I don't want my name on the publication only to have someone point out that there are more isometries that I didn't account for.

If you want to learn about this (not especially relevant to data science), you can refer to my Master's thesis: "Tractor Connections for Killing Tensors and their

Generalizations."

```
M > DGEEnvironment['VectorSpace'](6,V);  
Vector Space: V (16)
```

```
V > DGEEnvironment[VectorBundle](M,V,N);  
Vector Bundle: N (17)
```

```
N > G:=subs(M=N,g);  
 $G := (4x^2 + 2y^2 + 2z^2) dx \otimes dx + 2yx dx \otimes dy + 2zx dx \otimes dz + 2yx dy \otimes dx +$   
 $(2x^2 + 4y^2 + 2z^2) dy \otimes dy + 2yz dy \otimes dz + 2zx dz \otimes dx + 2yz dz \otimes dy +$   
 $(2x^2 + 2y^2 + 4z^2) dz \otimes dz$  (18)
```

```
N > Gamma:=Christoffel(G);
```

```
N > C:=TractorConnection[KillingVector](Gamma,N);
```

```
N > K:=CurvatureTensor(C);
```

```
N > K01:=ContractIndices(K,evalDG(D_x &w D_y),[[3,1],[4,2]]);
```

```
N > K02:=ContractIndices(K,evalDG(D_x &w D_z),[[3,1],[4,2]]);
```

```
N > K03:=ContractIndices(K,evalDG(D_y &w D_z),[[3,1],[4,2]]);
```

```
N > #Kmat:=Matrix([[convert(K01,DGMatrix)],[convert(K01,DGMatrix)  
],[convert(K01,DGMatrix)])]
```

```
N > IT0:=LieAlgebras:-InvariantTensors([convert(K01,DGMatrix),  
convert(K01,DGMatrix),convert(K01,DGMatrix)],DGinformation(N,  
"FrameFiberVectors"))
```

$$IT0 := \left[E1 - \frac{x}{z} E3 - \frac{2}{z} E5, E2 - \frac{y}{z} E3 - \frac{2}{z} E6, E4 - \frac{y}{z} E5 + \frac{x}{z} E6 \right] \quad (19)$$

```
N > nops(IT0)  
3 (20)
```

(This means there's an upper bound of 3 killing vectors. On the other hand, we found 3, so we know exactly what the Killing vectors are for this metric. It turns out they are the 3-d rotations. That was not planned, by the way.)

```
N > KV:=KillingVectors(G);
```

$$KV := \left[-\frac{z}{2} \partial_y + \frac{y}{2} \partial_z, -\frac{z}{2} \partial_x + \frac{x}{2} \partial_z, -\frac{y}{2} \partial_x + \frac{x}{2} \partial_y \right] \quad (21)$$

Now that we know the Killing vectors, let's return to the classification task. In our feature space, the separating hyperplane will be:

```
N > plane:=x1+x4/2-x5/sqrt(2)+x6/2;  
 $plane := x1 + \frac{1}{2} x4 - \frac{1}{2} x5\sqrt{2} + \frac{1}{2} x6$  (22)
```

This is what logistic regression should give us:

N > Ef:=eval(sigmoid,z=plane);

$$Ef := \frac{1}{1 + e^{-x1 - \frac{1}{2}x4 + \frac{1}{2}x5\sqrt{2} - \frac{1}{2}x6}} \quad (23)$$

Then, we wish to use the feature map to write this in the original (x,y) coordinates.

N > eval(Ef,eqns);

$$\frac{1}{1 + e^{-x^2 - \frac{1}{2}y^2 + yz - \frac{1}{2}z^2}} \quad (24)$$

Let's just check that we got what we started with...

N > simplify(eval(Ef,eqns)-f);

$$0 \quad (25)$$

Next, we construct an arbitrary linear combination of the Killing vectors:

N > X:=evalDG(a*KV[1]+b*KV[2]+c*KV[3]);

$$X := -\left(\frac{bz}{2} + \frac{cy}{2}\right)\partial_x - \left(\frac{az}{2} - \frac{cx}{2}\right)\partial_y + \left(\frac{ay}{2} + \frac{bx}{2}\right)\partial_z \quad (26)$$

Then we apply the linear combination to the function f and optimize the coefficients.

N > eqn1:=LieDerivative(X,f);

$$\begin{aligned} eqn1 := & \frac{2\left(-\frac{1}{2}bz - \frac{1}{2}cy\right)xe^{-x^2 - \frac{1}{2}y^2 + yz - \frac{1}{2}z^2}}{\left(1 + e^{-x^2 - \frac{1}{2}y^2 + yz - \frac{1}{2}z^2}\right)^2} \\ & - \frac{\left(-\frac{1}{2}az + \frac{1}{2}cx\right)(-y+z)e^{-x^2 - \frac{1}{2}y^2 + yz - \frac{1}{2}z^2}}{\left(1 + e^{-x^2 - \frac{1}{2}y^2 + yz - \frac{1}{2}z^2}\right)^2} \\ & - \frac{\left(\frac{1}{2}ay + \frac{1}{2}bx\right)(y-z)e^{-x^2 - \frac{1}{2}y^2 + yz - \frac{1}{2}z^2}}{\left(1 + e^{-x^2 - \frac{1}{2}y^2 + yz - \frac{1}{2}z^2}\right)^2} \end{aligned} \quad (27)$$

N > eqn2:=factor(eqn1);

(28)

$$eqn2 := -\frac{1}{2} \frac{e^{-x^2 - \frac{1}{2}y^2 + yz - \frac{1}{2}z^2} (y+z) (ay - az + bx + cx)}{\left(1 + e^{-x^2 - \frac{1}{2}y^2 + yz - \frac{1}{2}z^2}\right)^2} \quad (28)$$

$$\begin{aligned} \text{N} > eqn3 := \text{collect}(\text{op}(4, \text{numer}(eqn2)), \{x, y, z\}); \\ eqn3 &:= (b+c)x + ay - az \end{aligned} \quad (29)$$

$$\begin{aligned} \text{N} > eqns1 := [b+c, a]; \\ eqns1 &:= [b+c, a] \end{aligned} \quad (30)$$

$$\begin{aligned} \text{N} > sol := \text{solve}(eqns1) \\ sol &:= \{a=0, b=-c, c=c\} \end{aligned} \quad (31)$$

Here is a ground truth solution. The one we find using data will be a scaling of this (if we did everything right).

$$\begin{aligned} \text{N} > eX := \text{eval}(\text{eval}(X, sol), c=1); \\ eX &:= -\left(-\frac{z}{2} + \frac{y}{2}\right) \partial_x + \frac{x}{2} \partial_y - \frac{x}{2} \partial_z \end{aligned} \quad (32)$$